

Sai Varshith Mechineni

AI/ML ENGINEER

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SUMMARY

- **AI/ML Engineer** with 4+ years of experience in designing AI/ML solutions, applying **Python**, **Hugging Face Transformers**, and **FastAPI** for NLP-driven healthcare analytics to accelerate decision-making and reduce manual review times.
- Skilled in deploying semantic search frameworks with **Vector Databases** and **AWS SageMaker**, enhancing clinical insights through optimized embedding retrieval, and improving disease detection accuracy across large-scale unstructured datasets.
- Proficient in orchestrating cloud-native model deployments using **Docker**, **Kubernetes**, and **GitHub Actions**, ensuring automated testing and rapid production release cycles for enterprise-grade AI applications.
- Experienced in processing high-volume data pipelines with **Apache Spark**, **Kafka**, and **GCP Vertex AI**, delivering real-time and batch processing capabilities for multi-source data synchronization.
- Adept at implementing advanced model tuning and experiment tracking via **MLflow**, **Bayesian Optimization**, and **Boosting algorithms**, improving predictive accuracy in environmental and healthcare domains.
- Strong visualization and reporting expertise with **Seaborn**, **Plotly**, and **Power BI**, enabling actionable insights from complex geospatial, clinical, and environmental datasets for global stakeholders.

SKILLS

Languages:	Python, SQL, R
ML & Deep Learning:	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers, LangChain, OpenCV, NLP, Vector Databases
Generative AI & NLP:	GPT-4, BERT, LLaMA, LangGraph, RAG, Prompt Engineering, Text Generation, Tokenization
MLOps & Deployment:	MLflow, Docker, Kubernetes, FastAPI, GitHub Actions, CI/CD pipelines
Cloud Platforms:	AWS (SageMaker, Lambda, EC2), Azure (ML Studio, Databricks)
Data Analytics & Visualization:	Pandas, NumPy, Feature Engineering, Data Cleaning, Matplotlib, Seaborn, Plotly, Statsmodels, Normalization
Big Data & Pipelines:	Apache Spark, Kafka, Airflow
Model Evaluation:	Cross-validation, Precision, Recall, F1 Score, A/B Testing, ROC-AUC
Other Skills:	Hyperparameter tuning, Grid search, Bayesian Optimization, Object detection, Semantic segmentation, OOP principles, Multi-threading, Design patterns, Agile/Scrum methodologies

WORK EXPERIENCE

Johnson & Johnson, NJ, USA	Aug 2023 – Present
AI/ML Engineer	
• Designed and implemented machine learning models using Python, TensorFlow, and Random Forest to predict patient response and disease progression in oncology and immunology clinical trials. • Leveraged de-identified patient data and genomic profiles aggregated from Johnson & Johnson’s R&D data platform, enabling automated stratification of patient cohorts for precision medicine studies. • Applied Bayesian Optimization and Monte Carlo simulations to evaluate alternative trial designs and treatment arms across large patient cohorts. Improved outcome prediction precision and generated actionable insights into therapy effectiveness, accelerating real-world evidence generation for regulatory submissions. • Built a hybrid optimization framework combining Genetic Algorithms for adaptive cohort selection and Reinforcement Learning for optimizing longitudinal treatment strategies. Integrated longitudinal clinical trial records	

and biomarker data for continuous learning, improving trial efficiency and treatment response prediction accuracy by **22%**.

- Applied **BERT-based embeddings to encode unstructured clinical trial text data** (physician notes, adverse event reports, lab summaries, and drug safety narratives), enriching ML models with contextual insights for pharmacovigilance and safety monitoring.
- **Deployed end-to-end ML pipelines on AWS SageMaker** using SageMaker Pipelines for automated retraining, SageMaker Endpoints for real-time patient risk stratification, and CloudWatch for production monitoring and alerting. Ensured scalability, reproducibility, and compliance with FDA/EMA healthcare data standards.
- Optimized **SQL queries and ETL workflows** on Johnson & Johnson's enterprise-scale clinical and R&D data lake. Reduced data preparation and training time by **35%**, accelerating insights delivery for research scientists, biostatisticians, and clinical operations teams.
- Collaborated **with clinical researchers, pharmacovigilance teams, and medical affairs** to develop AI-powered dashboards in Power BI, visualizing trial efficacy metrics, safety signals, and therapy outcomes across subpopulations. Enabled data-driven resource allocation, improved patient safety, and supported evidence-based decision-making across global health systems.

Cognizant, India
2022

Apr 2020 – Aug

AI/ML Engineer

- Implemented machine learning models using **Scikit-learn, XGBoost, and LightGBM** to predict credit default probability, loan delinquency, and fraudulent insurance claims, enhancing automated underwriting decisions and reducing reliance on manual checks.
- Processed and transformed large-scale financial, credit bureau, and customer behavioral data using **Pandas and NumPy**, engineering features to capture payment patterns, credit utilization, delinquency history, and spending trends.
- Built automated **ETL pipelines in Python and SQL** to extract and cleanse data from core banking systems, credit bureaus, and CRM platforms, reducing data preparation time by **40%** and generating standardized, model-ready datasets that powered downstream analytics and machine learning.
- Used **LangChain** chains to automate customer communication generation (adverse action letters, fraud investigation summaries) with contextual accuracy based on model outputs and customer interaction history.
- Implemented **Apache Airflow** workflows to schedule model retraining, data refresh, and batch inference jobs, ensuring that ML predictions remained accurate with evolving customer behavior and changing regulatory requirements.
- Designed interactive **Tableau dashboards** for credit risk analysts, fraud teams, and loan officers to monitor key performance indicators such as loan approval rates, default probabilities, and fraud detection rates across customer segments, resulting in a **18% improvement** in decision-making speed and reporting efficiency.
- Employed **Azure Databricks** for large-scale data preprocessing, feature engineering, and distributed model training on financial, credit bureau, and behavioral datasets, improving data pipeline scalability and runtime efficiency.
- Applied custom **tokenization pipelines using spaCy and NLTK** for early-stage exploratory data analysis of customer logs—tokenized phrases were used to compute frequency-based fraud indicators (recurring complaints, refund requests).
- Built a **model performance monitoring system with Python** to track accuracy, drift, and alert thresholds, reducing false positives by **20%** and improving early warning signals for both credit and fraud detection models.
- Collaborated cross-functionally **with data engineers, compliance officers, and risk analytics teams** to embed models into credit origination, collections, and fraud triage systems, reducing the end-to-end loan decision cycle time by **25%**.
- Deployed custom fine-tuned **Transformers via Hugging Face transformers** and datasets libraries on **SageMaker** endpoints, enabling real-time NLP inference for customer risk profiling and regulatory checks.

EDUCATION

Master of Science in Information Technology – University of Cincinnati, Ohio, USA

Bachelor of Technology in Electronics and Communication Engineering – VIT, Amaravati, India